

**Sample Exam - Duration: 60 minutes**

Notes:

- **Closed-book** exam
- In each question there is only **one** correct answer
- Each group of **three wrong answers** cancels one correct answer
- Minimum grade: 7.5/20

Nº: \_\_\_\_\_ Name: \_\_\_\_\_

**1) One of the most relevant structural effects of adopting the RISC philosophy in a processor datapath is:**

- (a) The need for highly specialized functional units.
- (b) The reduction of the number of cycles per instruction through microcode.
- (c) The standardization of instruction formats, simplifying hardware control.
- (d) The increase in the number of supported addressing modes.

**2) In an ideal pipelined processor, the average CPI (clock cycles per instruction) approaches 1 mainly because:**

- (a) The pipeline reduces the individual execution time (latency) of each instruction.
- (b) The pipeline allows the overlap of execution of multiple instructions.
- (c) The clock cycle time is dynamically adjusted for each instruction.
- (d) Instructions are executed out of order.

**3) Consider a 32-bit variable with the value 0xAABBCCDD, stored in a big-endian architecture. The base address of the variable is B. What value is stored at address B + 1??**

- (a) 0xDD
- (b) 0xCC
- (c) 0xBB
- (d) 0xAA

**4) Consider the following RV32I instructions:**

- 1) `add x5, x6, x7`
- 2) `ld x10, 8(x11)`
- 3) `sd x10, 16(x11)`
- 4) `slli x5, x5, 2`

**Which of the following statements about their instruction formats is the only correct one?**

- (a) Instructions 2) and 4) use the same instruction format (I-type).
- (b) Instruction 3) uses the B-type format, due to the need to compute an offset.
- (c) All instructions belong to the R-type format, since they operate on registers.
- (d) Instruction 1) is the only one that uses the S-type format, intended for arithmetic operations.

**5) The base + offset addressing mode in RISC-V is essential to allow:**

- (a) Efficient access to local variables and parameters stored on the stack.
- (b) Direct access to absolute memory addresses.
- (c) Execution of PC-relative branches.
- (d) Elimination of the use of the stack pointer (SP).

**6) A data hazard of type WAR (Write After Read):**

- (a) Is eliminated through data forwarding techniques.
- (b) Only constitutes a real hazard in architectures that allow out-of-order execution.
- (c) Occurs frequently in classic in-order pipelines, requiring stalls.
- (d) Manifests exclusively in operations involving main memory accesses.

**7) Even in the presence of forwarding, a stall is unavoidable when:**

- (a) An arithmetic instruction depends on the result of another in the EX stage.
- (b) A value loaded from memory is used immediately by the following instruction.
- (c) Two instructions write to the same register.
- (d) A logical instruction follows an arithmetic instruction.

**8) In a deep pipeline, the main reason for the increased penalty associated with branch mispredictions is:**

- (a) Increased memory access time.
- (b) A larger number of speculatively executed instructions in the pipeline.
- (c) Reduced clock frequency.
- (d) Elimination of instruction-level parallelism.

**9) In a multiple-issue processor with dynamic scheduling, the exploitation of parallelism depends mainly on:**

- (a) Decisions taken at compile time.
- (b) Dynamic dependency analysis at run time.
- (c) Operating system intervention.
- (d) Increased memory bus bandwidth.

**10) To reduce the penalty of a branch misprediction, while keeping the predictor unchanged, an effective solution is to:**

- (a) Increase the size of the data cache.
- (b) Introduce out-of-order execution.
- (c) Resolve the branch earlier in the pipeline.
- (d) Increase the number of pipeline stages (deeper pipeline).

**11) In a classic 5-stage pipeline (IF, ID, EX, MEM, WB), if a branch instruction is resolved in the MEM stage, a misprediction implies:**

- (a) Flushing all instructions younger than the branch in the IF, ID and EX stages.
- (b) Only correcting the PC, without discarding any instructions.
- (c) Restarting the entire program execution.
- (d) Invalidating only the IF stage.

**12) Out-of-order execution improves performance mainly because it:**

- (a) Eliminates control dependencies.
- (b) Allows independent instructions to execute during stalls.
- (c) Reduces pipeline complexity.
- (d) Guarantees a constant CPI for any program.

**13) Which of the following code transformations is most effective in mitigating control hazards?**

- (a) Systematic insertion of *nop* instructions.
- (b) Reordering independent instructions after a branch.
- (c) Increasing the number of unconditional branches.
- (d) Replacing branches with memory accesses.

**14) The main impact of increasing cache associativity is:**

- (a) Reduction of compulsory misses.
- (b) Decrease in cache access latency.
- (c) Reduction of conflict misses.
- (d) Elimination of the need for block replacement.

**15) Keeping the total cache size constant, the use of larger blocks tends to:**

- (a) Increase tag overhead.
- (b) Reduce the exploitation of spatial locality.
- (c) Increase the probability of capacity misses.
- (d) Decrease the total number of cache blocks.

**16) In multiprocessor systems, a write-back cache:**

- (a) Simplifies coherence protocols.
- (b) Requires additional mechanisms to guarantee coherence between caches.
- (c) Eliminates the need for block invalidation.
- (d) Prevents concurrent access to main memory.

**17) Between two programs with the same number of instructions, the one with the higher average CPI is typically the one that:**

- (a) Contains more simple arithmetic instructions.
- (b) Executes more independent operations.
- (c) Suffers more pipeline flow interruptions.
- (d) Uses registers exclusively.

**18) In a RISC-V single-cycle processor, if we decide to implement the multiplication instruction (`mul`) using a functional unit that takes three times longer than the ALU, what is the direct impact on overall performance?**

- (a) CPI increases proportionally to the latency of the new unit.
- (b) Execution time of all programs increases, even those that do not use the `mul` instruction.
- (c) Only programs that use the `mul` instruction suffer a reduction in clock frequency..
- (d) Performance improves due to datapath specialisation.

**19) In the datapath of a RISC-V processor, what is the role of the PC and the ALU during the execution of a conditional branch instruction (e.g., `beq`)?**

- (a) The PC provides the address of the next instruction (PC+4), while the ALU computes only the branch target address.
- (b) The ALU decides whether the branch is taken and simultaneously computes the next PC value.
- (c) The ALU performs the register comparison, and dedicated add hardware computes the branch target address based on the PC and the immediate value.
- (d) The PC is incremented only after the ALU confirms that the branch is not taken.

**20) Consider a two-level cache hierarchy (L1 and L2). If we significantly increase the size of the L2 cache while keeping L1 unchanged, the most likely impact on the AMAT (Average Memory Access Time) observed by the CPU is:**

- (a) Reduction of the L1 hit time.
- (b) Reduction of the overall miss cost through a decrease in the L1 miss penalty.
- (c) Increase in compulsory misses in L2.
- (d) Complete elimination of accesses to main memory.